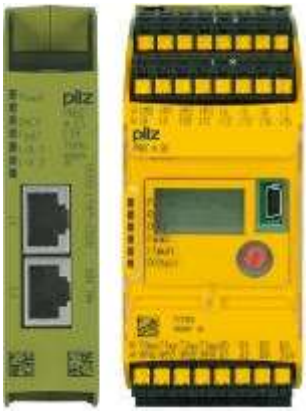
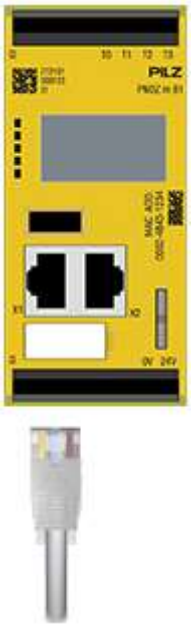
2.4 Communication via Modbus/TCP

For data exchange with Modbus/TCP, the PNOZmulti 2 is the connection server. All the service data is defined in one data record, to which the client has direct access.

Communication with Modbus/TCP is described in detail in the chapter entitled Modbus/TCP [72].

The following device combinations are possible:

Communication		Base unit
Communication Module PNOZ m ES ETH [53]		PNOZ m B0
Integrated Ethernet interface [54]		PNOZ m B1 from Version 1.1

INFORMATION

If virtual inputs/outputs are to be downloaded via the communication module (without using a fieldbus module), the interface "Inputs/outputs that are downloaded via the integrated interface" must be configured in the hardware configuration in the PNOZmulti Configurator.

3 Safety

3.1 Intended use

The communication interface on the configurable control system PNOZmulti 2 is used to transfer diagnostic data to an application program. This data may only be used for non-safety purposes, e.g. visualisation.



NOTICE

For details of the intended use and application of the configurable control system PNOZmulti 2 please refer to the operating instructions for the respective unit.

The following is deemed improper use in particular:

- Any component, technical or electrical modification to the product
- Use of the product outside the areas described in this manual
- Use of the product outside the technical details (see chapter entitled "Technical details").

3.2 Safety regulations

3.2.1 Use of qualified personnel

The products may only be assembled, installed, programmed, commissioned, operated, maintained and decommissioned by persons who are competent to do so.

A competent person is a qualified and knowledgeable person who, because of their training, experience and current professional activity, has the specialist knowledge required. In order to inspect, assess and handle products, devices, systems, plant and machinery, this person must be familiar with the state of the art and the applicable national, European and international laws, directives and standards.

It is the company's responsibility only to employ personnel who

- Are familiar with the basic regulations concerning health and safety / accident prevention,
- Have read and understood the information provided in the section entitled Safety
- Have a good knowledge of the generic and specialist standards applicable to the specific application.

3.2.2 Warranty and liability

All claims to warranty and liability will be rendered invalid if

- The product was used contrary to the purpose for which it is intended,
- Damage can be attributed to not having followed the guidelines in the manual,
- Operating personnel are not suitably qualified,
- Any type of modification has been made (e.g. exchanging components on the PCB boards, soldering work etc.).

3.2.3 Disposal

- When decommissioning, please comply with local regulations regarding the disposal of electronic devices (e.g. Electrical and Electronic Equipment Act).

4 Fieldbus Modules

4.1 Basics

The input and output range is each reserved an area of 32 Bytes for communication via fieldbuses; this is updated approx. every 15 ms. The Master (PC, PLC) can send 32 Bytes to the PNOZmulti 2 and receive 32 Bytes from the PNOZmulti 2. The Master can process the information in bytes, words or in double words.

4.1.1 Input data (to the PNOZmulti 2)

Double word	Word	Byte	Contents
0	0	0	State of virtual inputs on the PNOZmulti 2 fieldbus module. The inputs are defined in the PNOZmulti Configurator. Each input that is used is given a number there, e.g. i0, i5... The state of input i0 is stored in bit 0 of byte 0; the state of input i5 is stored in bit 5 of byte 0 etc.
		1	
	1	2	
		3	
1	2	4	
		5	
	3	6	
		7	
2	4	8	
		9	
	5	10	
		11	
3	6	12	
		13	
	7	14	
		15	
4	8	16	Reserved
		17	Table number
	9	18	Segment number
		19	Reserved
5	10	20	Reserved
		21	Reserved
	11	22	Reserved
		23	Reserved
6	12	24	Reserved
		25	Reserved
	13	26	Reserved
		27	Reserved

Double word	Word	Byte	Contents
7	14	28	Reserved
		29	Reserved
	15	30	Reserved
		31	Reserved

4.1.2 Output data (from the PNOZmulti 2)

Double Word	Word	Byte	Content
0	0	0	State of virtual outputs on the PNOZmulti 2 fieldbus module. The outputs are defined in the PNOZmulti Configurator. Each output that is used is given a number there, e.g. o0, o5... The status of output o0 is stored in bit 0 of byte 0; the status of output o5 is stored in bit 5 of byte 0 etc.
		1	
	1	2	
		3	
1	2	4	
		5	
	3	6	
		7	
2	4	8	
		9	
	5	10	
		11	
3	6	12	
		13	
	7	14	
		15	
4	8	16	LED status
		17	Table number
	9	18	Segment number
		19	Payload Byte 0
5	10	20	Payload Byte 1
		21	Payload Byte 2
	11	22	Payload Byte 3
		23	Payload Byte 4
6	12	24	Payload Byte 5
		25	Payload Byte 6
	13	26	Payload Byte 7
		27	Payload Byte 8

Double Word	Word	Byte	Content
7	14	28	Payload Byte 9
		29	Payload Byte 10
	15	30	Payload Byte 11
		31	Payload Byte 12

4.1.3 Overview of tables

The following types are available:

Table 20: Process data, base unit
Table 21: Process data, right-hand expansion modules
Table 22: Process data, left-hand expansion modules
Table 23: Process data, fieldbus module / communication module
Table 24: Process data, ST expansion module
Table 70: Diagnostic words
Table 71: Enable elements
Table 80: Project Information
Table 90: Device data, base unit
Table 91: Device data, right-hand expansion modules
Table 92: Device data, left-hand expansion modules
Table 93: Device data, fieldbus module / communication module
Table 94: Device data, ST expansion module
Table 110: Connected devices
Table 201: Advanced data

The content and the version and device-dependent validity of the tables are described in detail in the chapter Service data [86].

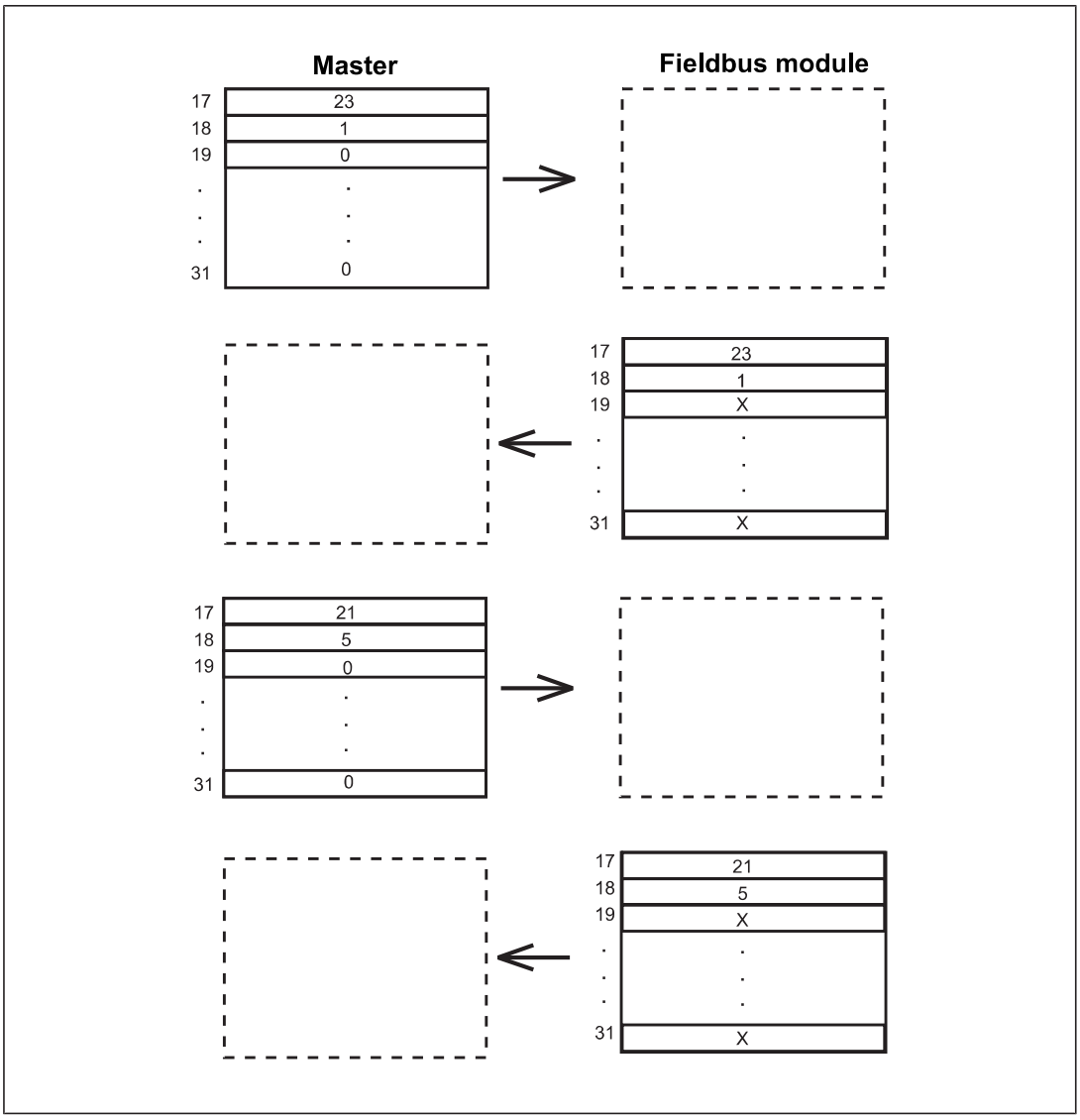
4.1.4 Access to table segments

Each table consists of one or more segments. Each segment is made up of 13 Bytes. The tables have a fixed assignment. The Master requests the required data using the table number and segment number. The Slave repeats the two numbers and sends the requested data. If data is requested that is not available, the Slave sends the error message FF (hex) or 255 (dec) instead of the segment number. The segments may be requested in any sequence.

4.1.4.1 Example 1 - Access successful

Access to existing segments

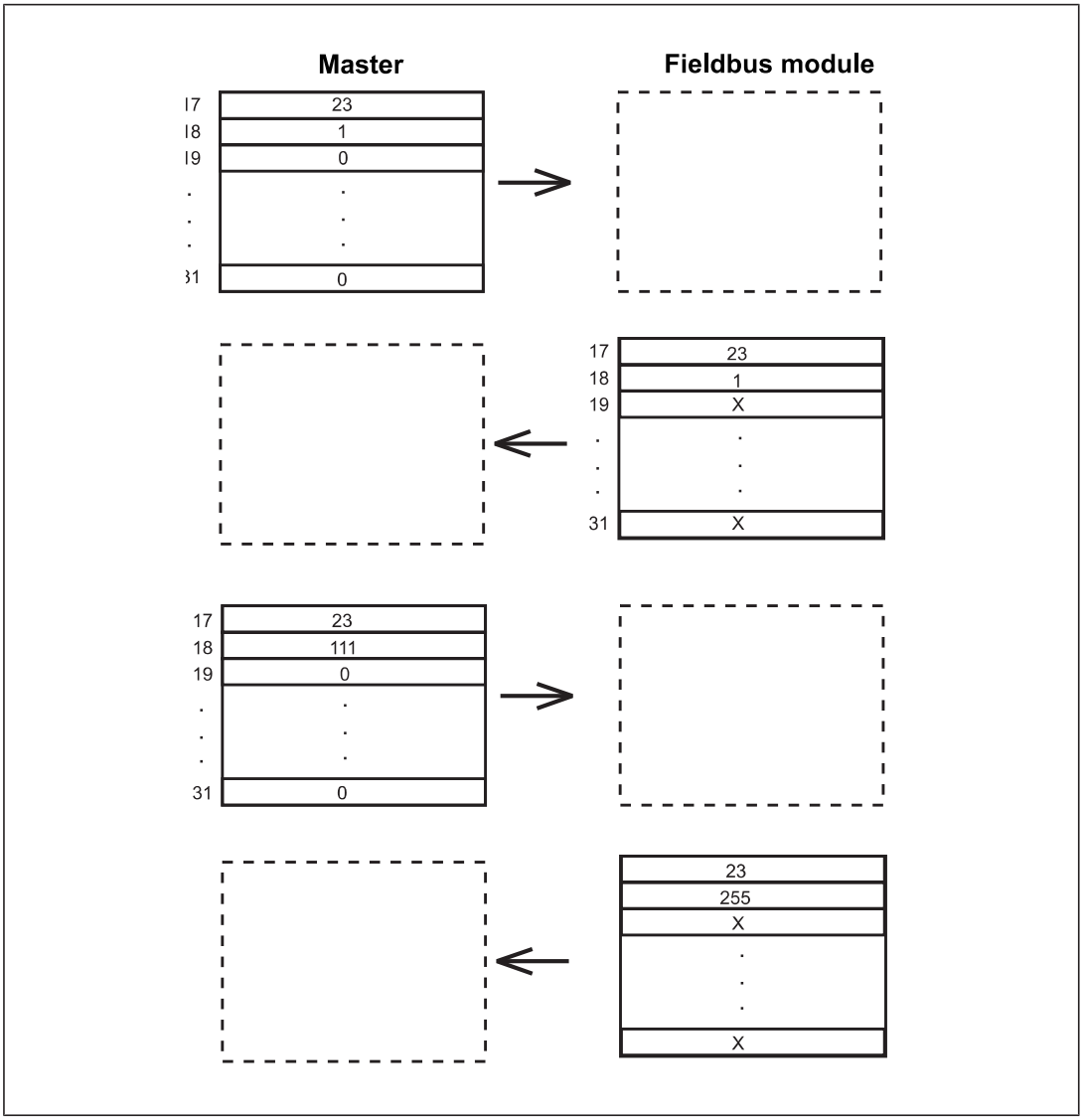
The Master requests segment 1 from table 23. The fieldbus module confirms both these details and sends segment 1. Then the data from segment 5 table 21 is requested.



4.1.4.2 Example 2 - Access failed

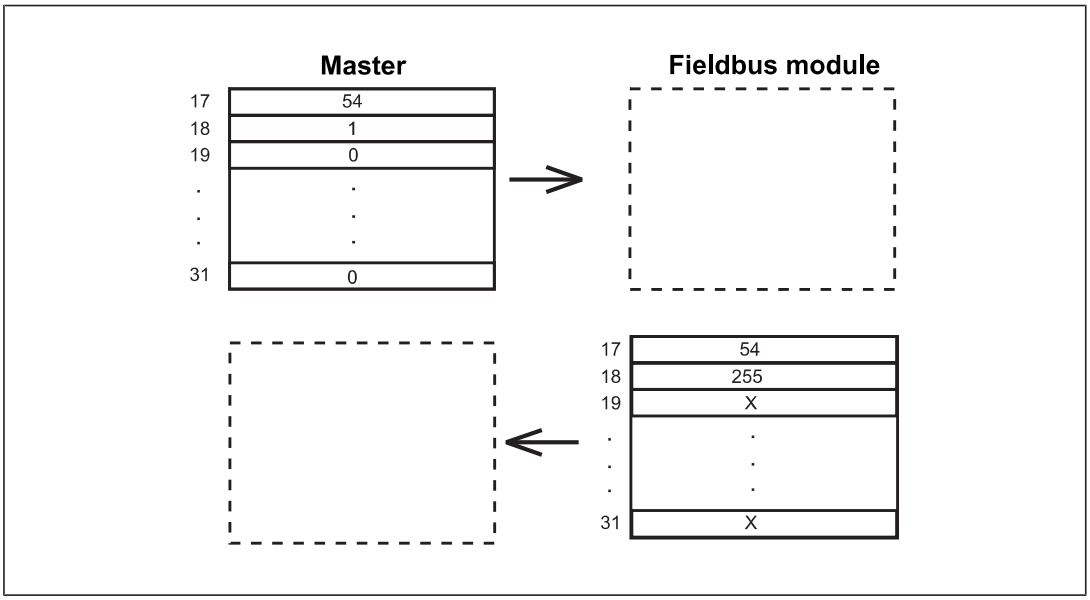
Access to non-existent segment

The Master requests segment 1 from table 23. The fieldbus module confirms both these details and sends segment 1. Then the Master requests segment 111 from table 23. As this table does not contain a segment 111, the Slave registers an error by sending back "255".



Access to non-existent table

The Master requests segment 1 from table 54. As there is no table 54, the Slave registers an error by sending back "255".



4.2 PNOZ m ES Profibus

4.2.1 Cyclical input and output data

Virtual inputs and outputs can be requested or set directly via the following modules. Each element can be selected individually in the master control system, e.g. virtual inputs i0-31. The data width is also established this way.

Input data

The Master writes to the virtual inputs of the PNOZmulti 2.

Description	Input data from PNOZmulti 2
Virtual inputs i0 – i31	4 Input Bytes
Virtual inputs i32 – i63	4 Input Bytes
Virtual inputs i64 – i95	4 Input Bytes
Virtual inputs i96 – i127	4 Input Bytes

Output data

The Master reads the virtual outputs of the PNOZmulti 2.

Description	Output data from PNOZmulti 2
Virtual outputs o0 – o31	4 Output Bytes
Virtual outputs o32 – o63	4 Output Bytes
Virtual outputs o64 – o95	4 Output Bytes
Virtual outputs o96 – o127	4 Output Bytes